Boss Challenge — Paper 36 (Class 7)

Nutrition in Animals | Heat | Algebraic Expressions | The Triangle & its Properties

One correct option each. Marking: +4 correct, -1 wrong, 0 blank. Write A/B/C/D on a separate sheet.

---

1. The breaking down of the food we eat into simpler substances the body can absorb is called:

(A) digestion
(B) absorption
(C) egestion
(D) respiration

2. A reliable measure of how hot or cold an object is, is given by its:

(A) length
(B) weight
(C) temperature
(D) colour

3. An expression that is built from variables and numbers joined by operations, such as $3x + 5$, is called an:

(A) equation
(B) fraction
(C) number line
(D) algebraic expression

4. A closed figure made by three line segments joined end to end is called a:

(A) circle
(B) triangle
(C) rectangle
(D) square

5. The set of organs through which food passes, from the mouth to the anus, is together called the:

(A) blood vessel
(B) alimentary canal
(C) wind pipe
(D) spinal cord

6. The instrument used to measure the temperature of a body is the:

(A) weighing scale
(B) thermometer
(C) barometer
(D) speedometer

7. In the term $7x$, the fixed number 7 multiplying the variable x is called the:

(A) variable
(B) constant
(C) coefficient
(D) exponent

8. A triangle has three sides, three corners and three:

(A) circles
(B) diagonals
(C) angles
(D) curves

9. The hard, white structures in the mouth that cut, tear and grind food are the:

(A) tonsils
(B) taste buds
(C) lips
(D) teeth

10. The special thermometer used to measure the temperature of the human body is the:

(A) clinical thermometer
(B) laboratory thermometer
(C) maximum thermometer
(D) barometer

11. In the expression $4y + 9$, the number 9 standing alone, with no variable attached, is called a:

(A) variable
(B) constant
(C) coefficient
(D) term with a variable

12. A triangle in which all three sides are equal in length is called an:

(A) isosceles triangle
(B) scalene triangle
(C) right triangle
(D) equilateral triangle

13. The watery liquid released in the mouth that wets food and begins to break down starch is:

(A) sweat
(B) saliva
(C) bile
(D) blood

14. The normal temperature of a healthy human body is taken to be about:

(A) 0°C
(B) 100°C
(C) 50°C
(D) 37°C

15. Terms that have exactly the same variable raised to the same power, such as 3x and 8x, are called:

- **(A)** equations
- **(B)** unlike terms
- **(C)** constant terms
- **(D)** like terms

16. A triangle in which exactly two sides are equal in length is called an:

- **(A)** isosceles triangle
- **(B)** equilateral triangle
- **(C)** obtuse triangle
- **(D)** scalene triangle

17. The muscular organ that mixes food with saliva and pushes it for swallowing is the:

- **(A)** tongue
- **(B)** lung
- **(C)** kidney
- **(D)** liver

18. Heat always flows on its own from a body at a higher temperature to one at a:

- **(A)** equal temperature
- **(B)** higher temperature
- **(C)** lower temperature
- **(D)** greater weight

19. The like terms 6a and 2a add together to give the single term:

- **(A)** 4a
- **(B)** 12a
- **(C)** 8a
- **(D)** $8a^2$

20. A triangle in which all three sides are of different lengths is called a:

- **(A)** equilateral triangle
- **(B)** scalene triangle
- **(C)** right triangle
- **(D)** isosceles triangle

21. After being swallowed, food is carried from the mouth down to the stomach through the:

- **(A)** wind pipe
- **(B)** food pipe (oesophagus)
- **(C)** small intestine
- **(D)** large intestine

22. The way heat travels through a solid, such as along a metal spoon in hot tea, mainly by passing from particle to particle, is called:

- **(A)** convection
- **(B)** reflection
- **(C)** radiation
- **(D)** conduction

23. The like terms 10m and 4m subtract to give:

- **(A)** 40m
- **(B)** 6m
- **(C)** 6
- **(D)** 14m

24. A triangle that has one of its angles exactly equal to 90° is called a:

- **(A)** equilateral triangle
- **(B)** right-angled triangle
- **(C)** acute-angled triangle
- **(D)** obtuse-angled triangle

25. The bag-like organ where food is churned with digestive juices and acid for a few hours is the:

- **(A)** stomach
- **(B)** heart
- **(C)** lung
- **(D)** brain

26. Heat moves through liquids and gases by the actual movement of the warmed fluid itself; this way of travelling is called:

- **(A)** conduction
- **(B)** radiation
- **(C)** freezing
- **(D)** convection

27. A pencil costs ■x. The cost of buying 5 such pencils is written as the expression:

- **(A)** x / 5
- **(B)** x + 5
- **(C)** x - 5
- **(D)** 5x

28. A triangle in which each of the three angles is less than 90° is called an:

- **(A)** acute-angled triangle
- **(B)** scalene triangle
- **(C)** obtuse-angled triangle
- **(D)** right-angled triangle

29. Most of the digested food is absorbed into the blood through the walls of the:

(A) small intestine
(B) stomach
(C) food pipe
(D) large intestine

30. Heat from the Sun reaches the Earth across empty space without any material in between, by:

(A) conduction
(B) convection
(C) radiation
(D) evaporation

31. A number is taken as n. The phrase '7 more than the number' is written as the expression:

(A) 7n
(B) n + 7
(C) n - 7
(D) 7 - n

32. A triangle in which one angle is greater than 90° is called an:

(A) obtuse-angled triangle
(B) right-angled triangle
(C) equilateral triangle
(D) acute-angled triangle

33. The tiny finger-like projections lining the small intestine that increase the surface for absorbing food are called:

(A) nephrons
(B) villi
(C) stomata
(D) alveoli

34. Materials such as metals that let heat pass through them easily are called good:

(A) insulators of heat
(B) conductors of heat
(C) absorbers of light
(D) reflectors of heat

35. If a number is p, then 'three times the number' is written as the expression:

(A) p + 3
(B) p / 3
(C) 3p
(D) p - 3

36. The three angles inside any triangle always add up to exactly:

(A) 270°

(B) 360°

(C) 180°

(D) 90°

37. The main job of the large intestine is to absorb water and some salts from the undigested:

(A) clean blood

(B) food waste

(C) fresh air

(D) chewed starch

38. Materials such as wood, plastic and wool that do not let heat pass through easily are called:

(A) insulators

(B) conductors

(C) radiators

(D) metals

39. The expression for 'a number x decreased by 4' is written as:

(A) 4 - x

(B) x + 4

(C) x - 4

(D) 4x

40. Two angles of a triangle are 50° and 60°. Using the angle-sum rule, the third angle is:

(A) 90°

(B) 110°

(C) 80°

(D) 70°

41. The semi-solid undigested waste is finally removed from the body through the:

(A) nose

(B) mouth

(C) skin

(D) anus

42. A handle made of wood or plastic is fixed to a metal cooking pan so that the cook's hand is protected, because such material is a poor:

(A) reflector of light

(B) insulator of heat

(C) conductor of heat

(D) source of heat

43. An expression that contains exactly one term, such as 5xy, is called a:

(A) monomial
(B) equation
(C) binomial
(D) trinomial

44. Each angle of an equilateral triangle measures, since the three equal angles share 180°:

(A) 30°
(B) 90°
(C) 45°
(D) 60°

45. The body's biggest gland, which makes bile to help in the digestion of fats, is the:

(A) pancreas
(B) kidney
(C) liver
(D) stomach

46. In hot sunny weather we are advised to wear light-coloured clothes because light colours:

(A) make their own heat
(B) reflect most of the heat
(C) block all the air
(D) absorb most of the heat

47. An expression made up of exactly two unlike terms, such as 3x + 5, is called a:

(A) monomial
(B) constant
(C) trinomial
(D) binomial

48. In a right-angled triangle, one angle is 90° and another is 40°. The third angle is:

(A) 40°
(B) 60°
(C) 130°
(D) 50°

49. Bile, which acts on fats during digestion, is made by the liver and stored in the:

(A) stomach
(B) gall bladder
(C) small intestine
(D) wind pipe

50. In cold winter weather we prefer dark-coloured clothes mainly because dark colours:

(A) let in more rain
(B) make their own heat
(C) absorb more heat
(D) reflect more heat

51. An expression containing exactly three terms, such as $x^2 + 3x + 2$, is called a:

(A) monomial
(B) binomial
(C) coefficient
(D) trinomial

52. A line drawn from one vertex of a triangle straight to the mid-point of the opposite side is called a:

(A) median
(B) altitude
(C) hypotenuse
(D) base

53. The flat, broad front teeth used mainly for biting and cutting food are called:

(A) canines
(B) molars
(C) premolars
(D) incisors

54. Wearing two thin layers of clothing in winter is often warmer than one thick layer because air trapped between the layers is a poor:

(A) source of heat
(B) good conductor of heat
(C) reflector of sound
(D) conductor of heat

55. When $x = 2$, the value of the expression $3x + 1$, found by putting 2 in place of x, is:

(A) 9
(B) 5
(C) 8
(D) 7

56. The perpendicular distance from a vertex of a triangle to the line of its opposite side is called the:

(A) altitude (height)
(B) hypotenuse
(C) median
(D) perimeter

57. The pointed teeth, next to the front teeth, that are used for tearing and piercing food are the:

(A) canines
(B) wisdom teeth
(C) molars
(D) incisors

58. On the Celsius scale, the temperature at which pure water freezes into ice is:

(A) 100°C
(B) 0°C
(C) 50°C
(D) 37°C

59. When y = 5, the value of the expression 2y - 3 is:

(A) 4
(B) 7
(C) 13
(D) 10

60. In a right-angled triangle, the longest side, lying opposite the right angle, is called the:

(A) altitude
(B) median
(C) hypotenuse
(D) base

61. Cattle such as cows quickly swallow grass, store it, and later bring it back to the mouth to chew again; this returned food is called:

(A) cud
(B) starch
(C) saliva
(D) bile

62. On the Celsius scale, the temperature at which pure water boils is:

(A) 37°C
(B) 0°C
(C) 100°C
(D) 50°C

63. Simplifying the expression 4x + 3x by combining the like terms gives:

(A) x
(B) 7x
(C) $7x^2$
(D) 12x

64. By the Pythagoras property, in a right triangle the square on the hypotenuse equals the sum of the squares on the other two:

(A) sides
(B) medians
(C) altitudes
(D) angles

65. Grass-eating animals like cows can digest the cellulose in grass with the help of certain microbes living in their:

- **(A)** brain
- **(B)** lungs
- **(C)** teeth
- **(D)** rumen (part of the stomach)

66. The sea breeze that blows from the sea toward the land during the daytime is a result of heat transfer by:

- **(A)** conduction
- **(B)** freezing
- **(C)** radiation
- **(D)** convection

67. In the expression $2x + 3y$, the terms $2x$ and $3y$ cannot be added into one term because they are:

- **(A)** unlike terms
- **(B)** constants
- **(C)** equal terms
- **(D)** like terms

68. A right triangle has legs of 3 cm and 4 cm. By Pythagoras, since $3^2 + 4^2 = 9 + 16 = 25$, the hypotenuse is:

- **(A)** 25 cm
- **(B)** 5 cm
- **(C)** 7 cm
- **(D)** 12 cm

69. The tiny single-celled animal Amoeba captures its food by pushing out finger-like extensions called:

- **(A)** tentacles
- **(B)** cilia
- **(C)** villi
- **(D)** pseudopodia

70. In a clinical thermometer, the silvery liquid that rises in the thin tube to show the temperature is traditionally:

- **(A)** mercury
- **(B)** oil
- **(C)** water
- **(D)** milk

71. A cow eats k kg of fodder each day. The fodder a herd of 6 such cows eats in one day is the expression:

- **(A)** $6k$
- **(B)** $k / 6$
- **(C)** $k + 6$
- **(D)** $k - 6$

72. The exterior angle of a triangle is equal to the sum of the two interior opposite angles. If those two are 45° and 65°, the exterior angle is:

- **(A)** 20°
- **(B)** 90°
- **(C)** 110°
- **(D)** 70°

73. Within an Amoeba's body, the captured food is digested inside a small bubble-like bag known as the:

- **(A)** food vacuole
- **(B)** gall bladder
- **(C)** villus
- **(D)** nucleus

74. A liquid in a thermometer rises up the tube when heated because, on heating, the liquid:

- **(A)** disappears
- **(B)** contracts
- **(C)** expands
- **(D)** freezes

75. A plant is now h cm tall and grows 2 cm taller. Its new height is written as the expression:

- **(A)** h + 2
- **(B)** 2 - h
- **(C)** 2h
- **(D)** h - 2

76. Adding the lengths of any two sides of a triangle always gives a total greater than the remaining:

- **(A)** area
- **(B)** third side
- **(C)** perimeter
- **(D)** longest angle

77. The young of a human baby first feed only on their mother's:

- **(A)** grass
- **(B)** bread
- **(C)** milk
- **(D)** raw rice

78. The temperature of a cup of water rises from 25°C to 70°C when heated. The rise in temperature, found by 70 - 25, is:

- **(A)** 95°C
- **(B)** 25°C
- **(C)** 45°C
- **(D)** 70°C

79. The breathing rate of a child rises from r breaths a minute by 5 breaths during play. The new rate is the expression:

- **(A)** 5r
- **(B)** r + 5
- **(C)** r - 5
- **(D)** 5 - r

80. Can a triangle be drawn with sides 2 cm, 3 cm and 8 cm? Because 2 + 3 = 5 is not greater than 8, such a triangle is:

- **(A)** possible and equilateral
- **(B)** not possible
- **(C)** possible and right-angled
- **(D)** possible and isosceles

81. A cow's small intestine is very long because such a long tube gives more time and surface to digest tough:

- **(A)** milk
- **(B)** meat
- **(C)** water
- **(D)** grass (plant food)

82. Tea cools from 80°C to 65°C while it stands. The fall in temperature, found by 80 - 65, is:

- **(A)** 80°C
- **(B)** 15°C
- **(C)** 65°C
- **(D)** 145°C

83. Each test tube holds t mL of liquid. The total liquid in 4 identical test tubes is the expression:

- **(A)** 4t
- **(B)** t - 4
- **(C)** t + 4
- **(D)** t / 4

84. An equilateral triangular sign has each side 9 cm long. Its perimeter, found by 3 x side, is:

- **(A)** 12 cm
- **(B)** 81 cm
- **(C)** 27 cm
- **(D)** 18 cm

85. A child has 8 incisor teeth, each about 6 mm wide. The total width of all the incisors set side by side, found by 8 x 6, is:

- **(A)** 2 mm
- **(B)** 40 mm
- **(C)** 14 mm
- **(D)** 48 mm

86. Two beakers of water are at 30°C and 50°C. Their average temperature, found by (30 + 50) / 2, is:

- **(A)** 15°C
- **(B)** 80°C
- **(C)** 40°C
- **(D)** 20°C

87. The number of terms in the expression $5x^2 + 2x + 7$ is:

- **(A)** three
- **(B)** five
- **(C)** one
- **(D)** two

88. A triangular plot has sides 12 m, 15 m and 13 m. The fencing needed once around it (its perimeter), found by 12 + 15 + 13, is:

- **(A)** 25 m
- **(B)** 180 m
- **(C)** 40 m
- **(D)** 27 m

89. A cow's small intestine is about 40 m long and its large intestine about 10 m. The total length of both, found by 40 + 10, is:

- **(A)** 400 m
- **(B)** 4 m
- **(C)** 50 m
- **(D)** 30 m

90. A doctor uses a clinical thermometer rather than a laboratory one for a patient because the clinical thermometer's scale is suited to the:

- **(A)** range up to boiling water
- **(B)** very wide range for hot experiments
- **(C)** narrow range around body temperature
- **(D)** freezing range below 0°C

91. When x = 3, the value of the expression x^2 (which means x x x) is:

- **(A)** 5
- **(B)** 23
- **(C)** 9
- **(D)** 6

92. An isosceles triangle has two equal angles. If its third (unequal) angle is 80°, then each of the two equal angles is:

- **(A)** 100°
- **(B)** 50°
- **(C)** 40°
- **(D)** 80°

93. The wave-like squeezing movement of the gut walls that pushes food forward through the food pipe and intestine is called:

- **(A)** respiration
- **(B)** absorption
- **(C)** digestion
- **(D)** peristalsis

94. On a hot day, a steel chair feels hotter to the touch than a wooden one kept beside it because steel is a better:

- **(A)** insulator of heat
- **(B)** reflector of heat
- **(C)** conductor of heat
- **(D)** source of heat

95. The like terms in the expression 8p + 3q - 2p are 8p and -2p, which combine to give:

- **(A)** 6p
- **(B)** 11p
- **(C)** 6pq
- **(D)** 9p

96. A ladder leaning on a wall makes a right-angled triangle with the ground. The right angle is formed between the wall and the:

- **(A)** ground
- **(B)** sky
- **(C)** sun
- **(D)** ladder

97. The throwing out of undigested food as waste from the body is called:

- **(A)** egestion
- **(B)** ingestion
- **(C)** absorption
- **(D)** digestion

98. When you hold an ice cube, your hand feels cold because heat flows out of your warm hand and into the:

- **(A)** hand from the ice
- **(B)** warmer air
- **(C)** sunlight
- **(D)** colder ice

99. Each empty beaker has a mass of b grams. The total mass of 3 such beakers, with an extra 50 g of water added, is the expression:

- **(A)** 50b + 3
- **(B)** 3b + 3
- **(C)** 3b - 50
- **(D)** 3b + 50

100. A triangle can never have two right angles, because two 90° angles already total 180°, leaving nothing for the:

- **(A)** perimeter
- **(B)** third angle
- **(C)** first angle
- **(D)** longest side